

Where Models Miss: A Cross-Platform Analysis of GUI Action Grounding

Aliaa Gheis^{*1} Menna Abdelbaset^{*1} Yara Hisham^{*1} Nouran Hany^{*1} Sarah Elzayat^{*1}

Abstract

1. Introduction

2. Related Work

3. Method

Click2Act is an open, extendable evaluation framework for GUI agents. It supports two evaluation modes—point grounding and action prediction—and is designed to be modular: new agents and benchmarks are registered independently and paired at runtime via YAML configuration files as illustrated in Figure 1.

3.1. Evaluated Models

We evaluate four small pure-vision VLMs, all used with no additional fine-tuning. Table 1 summarizes each model.

Table 1. Models evaluated in this study.

Model	Params
UI-AGILE-3B	3B
Holo2-4B	4B
AGUVIS-7B	7B
AGUVIS+UI-Zoomer	7B

All evaluated models fall in the 3B–7B parameter range, targeting practical deployment on consumer devices like laptops and workstations without cloud infrastructure. Despite this appeal, how much accuracy they sacrifice compared to flagships such as UI-Venus-1.5-30B-A3B (Venus-Team et al., 2026) and Holo2-235B-A22B (Company, 2025), and where exactly they fail, remains an open question.

AGUVIS-7B (Xu et al., 2024) is a fully open-source autonomous GUI agent built on Qwen2-VL that supports all

^{*}Equal contribution ¹Computer Engineering Department, Cairo University Faculty of Engineering, Egypt, Cairo. Correspondence to: Aliaa Gheis <aliaagheis@gmail.com>.

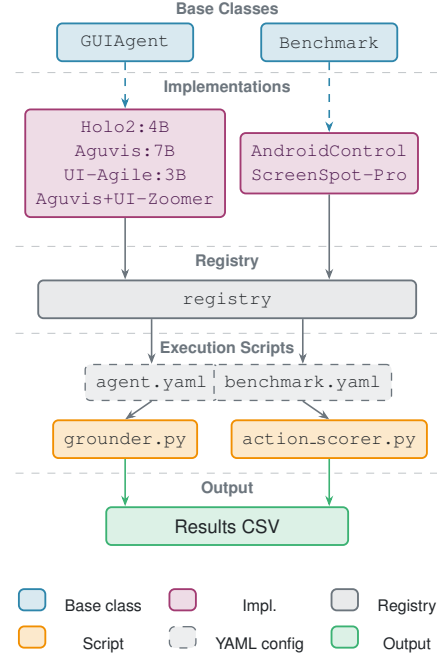


Figure 1. Click2Act framework overview. Abstract base classes (GUIAgent, Benchmark) are subclassed by concrete implementations and resolved at runtime via a registry. Each execution script receives an agent.yaml and a benchmark.yaml, runs inference over benchmark samples, and writes results to a CSV for offline analysis. Dashed arrows = inheritance; solid arrows = data flow.

platforms. It operates in two stages: (1) a planning stage where the model generates a structured *inner monologue*—consisting of a textual description of the current screen, internal reasoning toward the high-level goal, and low-level action instructions—and (2) a grounding stage where the inner monologue is translated into executable actions. It was one of the first small fully open-source vision-based GUI agents to demonstrate competitive performance on ScreenSpot.

Holo2-4B (Company, 2025) is a fully open-source autonomous GUI agent built on Qwen3-VL that supports all platforms. It is trained in two stages: (1) a large-scale supervised fine-tuning (SFT) stage on screen localization and element grounding, and (2) an online reinforcement learning

stage using Group Relative Policy Optimization (GRPO) to refine its multi-step reasoning and navigation policies.

UI-Agile-7B

AGUVIS+UI-Zoomer

3.2. Experimented Benchmarks

3.3. Evaluation Metrics

4. Experiments

5. Conclusion

Acknowledgements

Impact Statement

References

Company, H. Holo2 - open foundation models for navigation and computer use agents, 2025. URL <https://huggingface.co/collections/Hcompany/holo2>.

Li, W., Bishop, W., Li, A., Rawles, C., Campbell-Ajala, F., Tyamagundlu, D., and Riva, O. On the effects of data scale on computer control agents. *arXiv preprint arXiv:2406.03679*, 2024.

Venus-Team, Gao, C., Gu, Z., Liu, Y., Qiu, X., Shen, S., Wen, Y., Xia, T., Xu, Z., Zeng, Z., Zhou, B., Zhou, X., Chen, W., Dai, S., Dou, J., Gong, Y., Guo, Y., Guo, Z., Li, F., Li, Q., Lin, J., Zhou, Y., Zhu, L., Chen, L., Guo, Z., Meng, C., and Wang, W. Ui-venus-1.5 technical report, 2026. URL <https://arxiv.org/abs/2602.09082>.

Xu, Y., Wang, Z., Wang, J., Lu, D., Xie, T., Saha, A., Sahoo, D., Yu, T., and Xiong, C. Aguis: Unified pure vision agents for autonomous gui interaction, 2024.

(Xu et al., 2024) (Li et al., 2024)

A. Appendix